

Synthesis of p-CuO/n-ZnO heterostructure by microwave hydrothermal method and evaluation of its photo and bio-catalytic performance

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ABSTRACT

The use of photocatalysts without noble metals is of great interest in the industrial field for the degradation of organic pollutants. In this study, a CuO/ZnO heterostructure was synthesized using the microwave hydrothermal method and characterized using various analytical techniques. The synthesized CuO/ZnO photocatalyst exhibited a low bandgap energy of 2.4 eV, enabling efficient visible light absorption. The photocatalytic activity of the CuO/ZnO heterostructure was evaluated for the degradation of Methyl Orange (MO) dye and showed a high degradation efficiency of 99 % due to its excellent electron-hole charge separation. The biological activity of the synthesized CuO/ZnO catalyst was further investigated through protein docking studies, which showed promising results. The CuO/ZnO was also evaluated for its anticancer and antibacterial properties. It exhibited effective anticancer activity against prostate cancer cells (PC-3) in a dose-dependent manner, with an IC₅₀ value of 6.87 ± 8 . In addition, it demonstrated potent antibacterial activity against *Escherichia coli*, *Staphylococcus aureus*, *Bacillus cereus* and *Pseudomonas aeruginosa*. The results of this study demonstrate the potential of CuO/ZnO heterostructures as promising materials for various applications in the fields of photocatalysis, biomedicine and antimicrobial materials. Future research in this area will focus on further optimizing the properties of the CuO/ZnO heterostructure to enhance its performance in these applications.

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1. Introduction

As the world is facing a continuous rise in pollution and depletion of natural resources, environmental sustainability has become a crucial concern [1,2]. Photocatalysis is a promising technology that could provide a solution to environmental problems. It involves the use of a photocatalyst to catalyze a chemical reaction under light irradiation, which can effectively decompose pollutants into harmless substances [3–6].

Photocatalysis has been widely used in the treatment of water and air pollution, as well as in self-cleaning surfaces, antibacterial coatings, energy production and food industry [7–9]. The photocatalysis process is based on the principle of photoexcitation, which occurs when a photon of light is absorbed by a semiconductor material [10]. This excites an electron from its valence band to the conduction band, leaving behind a hole in the valence band [11]. These photo-generated electron-hole pairs can participate in redox reactions, leading to the degradation of pollutants [12]. Photocatalytic oxidation is a type of photocatalysis that involves the oxidation of organic and inorganic compounds in the presence of a photocatalyst and light [13]. The process is initiated by the adsorption of the pollutant on the surface of the photocatalyst, followed by its oxidation through the transfer of electrons and holes [14]. This results in the formation of reactive oxygen species, which can further react with the pollutant and degrade it into harmless substances [15,16].

CuO/ZnO p-n junction photocatalysts have attracted significant attention due to their unique properties and superior photocatalytic activity [17,18]. Several studies have investigated the synthesis and characterization of CuO/ZnO p-n junction photocatalysts using various methods [19], such as hydrothermal, sol-gel, and co-precipitation [20,21]. These studies have demonstrated that the morphology, size, and composition of the photocatalyst significantly influence its photocatalytic activity [22]. The characterization of the photocatalysts is typically performed using techniques such as X-ray diffraction, scanning electron microscopy, transmission electron microscopy, and UV-Vis spectroscopy [23,24]. These techniques provide information about the crystal structure, morphology, particle size, and optical properties of the photocatalyst. CuO/ZnO p-n junction photocatalysts [25] have been shown to enhance photocatalytic activity due to the synergetic effect between CuO and ZnO [26,27]. The formation of a p-n junction between CuO and ZnO leads to the separation of photo-generated electron-hole pairs, which reduces their recombination rate and increases the photocatalytic efficiency [28]. Additionally, the presence of CuO on the surface of ZnO can improve the adsorption of pollutants and facilitate their oxidation. Several factors can influence the performance of CuO/ZnO p-n junction photocatalysts, including the morphology, size, composition, and preparation method of the photocatalyst [20,29].

The type of pollutant, its concentration, and the pH of the solution can also affect the photocatalytic activity [7]. The light source, intensity, and wavelength can significantly impact the efficiency of photocatalysis [30]. Recent studies have focused on improving the performance of CuO/ZnO p-n junction photocatalysts by modifying their structure and composition. For example, the use of co-doping or surface modification techniques can enhance the separation of electron-hole pairs and improve the adsorption of pollutants [31]. Additionally, the integration of CuO/ZnO p-n junction photocatalysts with other materials, such as graphene or carbon nanotubes, can further enhance their photocatalytic activity [32,33]. CuO/ZnO p-n junction photocatalysts have been compared with other metal-based photocatalysts, such as TiO₂ and Ag doped ZnO [27]. While TiO₂ is the most studied and widely used photocatalyst [34–37], ZnO photocatalysts also exhibit high photocatalytic activity, but they suffer from poor stability and low selectivity [23]. The development of CuO/ZnO p-n junction photocatalysts is still at an early stage, and several challenges need to be addressed. The optimization of the synthesis and preparation methods, the understanding of the photocatalytic mechanism, and the improvement of their stability and selectivity are some of the challenges that need to be overcome [38]. However, with the continuous development of new materials and techniques, CuO/ZnO p-n junction photocatalysts have promising prospects in various applications related to environmental sustainability [39].

In this article, we have focused on the development of CuO/ZnO p-n junction photocatalysts for enhanced photocatalysis and biological applications [40]. The synthesis and characterization techniques, factors influencing their performance of photocatalysts have been discussed. Future research should focus on addressing the challenges and optimizing the performance of CuO/ZnO p-n junction photocatalysts for various applications related to environmental sustainability. Here in the CuO/ZnO composite has semiconductor properties with low bandgap and shows high efficiency towards degradation of dyes.

In the present work methyl orange (MO) were selected for the degradation, which are very toxic to living organisms. The CuO/ZnO composite has been used to analyze the bio-catalytic activity, our research involves a look at the CuO/ZnO nanostructures potential anticancer properties. Therapeutic approaches are the best option for treating cancer because they are more effective than other therapies (regular chemotherapy) and have fewer negative side effects, such as reduced viability, fewer adverse effects, lower therapeutic indicators, resistance to multiple drugs, and broader treatment goals [41–43]. *In vitro* cytotoxicity of CuO/ZnO nanoparticles was evaluated against prostate cancer (PC-3) cell lines and also assessed the anti-bacterial activity with various species. Many studies with the bactericidal effects of metal oxide nanoparticles were observed on Gram-positive and Gram-negative bacteria, as well as spores, which are microorganisms that are resistant to high temperatures and high pressures [42,44]. As a result, further clinical studies are needed to demonstrate the efficacy of CuO/ZnO nanostructures.

2. Materials and methods

The chemicals used in the present study were of analytical grade, ensuring high purity and minimizing the presence of impurities. Copper nitrate trihydrate extra pure (98.6 %, 8 Loba Chemie) was used as the source of copper, while Zinc nitrate (Fisher Scientific, 98 %) was used as the source of zinc. NaOH (Fisher Scientific, 98 %) was used as the base to precipitate the hydroxides of Copper and Zinc. To ensure the purity of the resulting product, Nano pure water (18.2 M cm) was used as the solvent. The composite preparation was

performed through microwave hydrothermal method using the Anton Parr Multi-wave Pro, which is a closed system designed for safe and efficient synthesis of materials under high-pressure and high-temperature conditions. The use of high-quality chemicals and state of the art equipment in the present study ensures the reliability and reproducibility of the results, and enhances our understanding of the properties and potential applications of the CuO/ZnO heterostructure synthesized by microwave hydrothermal method [23,42].

2.1. Synthesis of CuO/ZnO heterostructure

The synthesis of the CuO/ZnO heterostructure was carried out using the microwave hydrothermal method. To obtain 1 wt% CuO/ZnO composite, a mixture of 0.11 g of Cu (NO₃)₂•3H₂O stock solution and 2.26 g of Zn (NO₃)₂•6H₂O stock solution was prepared. Dropwise addition of 0.5 M NaOH solution to the mixed solution was performed with constant stirring until the pH reached 9.0–9.5. By varying the amounts of Cu (NO₃)₂•3H₂O stock solution, CuO/ZnO composites with Cu weight percentages of 2 %, 3 %, and 4 % in the final product were prepared, namely 1 wt% CuO/ZnO composite (CZ1), 2 wt% CuO/ZnO composite (CZ2), 3 wt% CuO/ZnO composite (CZ3), 4 wt% CuO/ZnO composite (CZ4). Resulting in the formation of a white-colored precipitate of hydroxides of Copper and Zinc. The resulting solution was transferred to a Teflon liner. The liner was placed inside the Microwave hydrothermal system, where it experienced a 20-min duration of elevated temperature, followed by holding the temperature for 30 min, with low stirring conditions. After the system cooled down, the obtained product was washed by centrifugation technique, then the product was lyophilized to obtain a pure product. The resulting CuO/ZnO heterostructure was suitable for further analysis by various analytical techniques to understand its properties and potential applications [23].

2.2. Characterization techniques

The CuO/ZnO nanocomposite was analyzed by several analytical techniques to understand its crystalline properties, surface morphology, elemental composition, functional groups, electrical behavior, and charge migration during photocatalysis.

XRD analysis was performed using a Rigaku Ultima III series TEX system with Cu K α (λ = 0.15405 nm) at 2°/min to understand the crystalline properties of the nanocomposite. SEM analysis was performed using a HITACHI S 3400 N, Japan, to study the surface morphology of the nanocomposite, while the EDX Hitachi system 2, USA, was used to determine the elements present in the photocatalyst. The functional groups in the nanocomposite were investigated by FTIR using JASCO 460 plus, Japan, in the range 1000–400/cm at 4/cm resolution. The UV–Vis spectrophotometer (DU739, Germany) was used to analyze the synthesized CuO/ZnO nanocomposite, while the reflectance of the near-infrared region was measured by spectroradiometer UV Vis NIR SWIR (RS3500) in the range 350–2500 nm. Raman spectroscopy was used to analyze the electrical behavior and sample resistance of the nanocomposite, while the elemental composition and defects were investigated by XPS spectrophotometer. Finally, PL spectroscopy was used to analyze the charge migration during photocatalysis using JASCO spectrophotometer (FP 8300). Transmission electron microscope (TEM)-Talos 200kv produced by Thermo Fisher Scientific. It operates at an accelerating voltage of 200 kV (kV) and is capable of achieving high-resolution imaging of samples with sub-nanometer resolution. These analytical techniques provided a comprehensive understanding of the CuO/ZnO nanocomposite synthesized by the microwave hydrothermal method, and helped to determine its suitability for photocatalysis.

2.3. Dye degradation

The study investigated the photocatalytic properties of CuO/ZnO heterostructure using a Mercury lamp as a visible light alternative. The degradation process was carried out by preparing a mixture of 100 ml of 10 mg L⁻¹ (10 ppm) MO dye and 0.05 g of CuO/ZnO photocatalyst. The mixture was stirred for 30 min to attain adsorption equilibrium. The Mercury lamp was then passed over the mixture, and 4 ml of the mixture was pipetted out at every 10 min. The samples were centrifuged and the absorbance was measured at λ -max. The percentage of photocatalytic degradation efficiency was measured using the formula % of Photodegradation efficiency = (Ci-Cf/Ci) x 100, where Ci is the initial concentration and Cf is the final concentration. Additionally, the total organic carbon (TOC) content of the degradation product was measured using a Shimadzu TOC-L series analyzer to estimate the total mineralization degree.

2.4. Electrochemical measurements

The electro chemical behavior and photocurrent properties as CuO/ZnO composite was investigated by electro chemical analyzer CH1608E work station, containing platinum electrode, Glass electrode and calomel electrode. Aqueous solution of 0.5 m Na₂SO₄ was used as electrolyte and mercury lamp with 400 W as visible light source. 4 mg of CuO/ZnO composite was dispersed in 20 μ L of 5 % Nafion solution and 1 ml of 3:7 ethanol and distilled water. The dispersed suspension was ultrasonicated for 30 min to get homogenous solution. The solution was coated on FTO Glass working electrode with 1 cm² area for the electro chemical experiment.

2.5. Molecular docking

In the current study, Chem Draw was used to construct a nanocomposite CuO/ZnO based on an experimental reaction. Using ChemBio 3D Ultra and following the established technique, nanocomposite's energy was minimized, and the nanocomposite was then stored as a PDB file. AutoDock was employed to establish the molecular interactions between nanocomposites and target protein. Visualizing the docking molecular interaction between ligands and proteins was made possible through the use of BIOVIA discovery.

2.6. Anti-cancer MTT assay

To determine prostate cancer (PC3) cell viability, the protocol of Alley et al. (1988) and slightly modified method of Sophia P et al. (2022) was used [45,46]. The MTT cell proliferation instruction guidelines were used. The cells were grown in T-25 flasks, harvested, and plated at a cell density of 10,000 cells per well in a 96 well plate in DMEM. After 24 h, the cells were treated with various concentrations of CuO/ZnO nanocomposite. After 24 h, the MTT reagent was added at a concentration of 0.5 mg/ml and incubated in the CO₂ incubator for 3 h. Excess MTT was removed from the media, and formazan crystals were dissolved in 100 μ l DMSO. The purple color formazan solution was measured at 570 nm, and the background reading was subtracted at 630 nm. The viability of PC3 cells was calculated as a percentage.

2.7. Anti-bacterial assay

Gentamicin and ciprofloxacin 10 μ g concentration was employed standard antibiotic, nanocomposites were dispersed in DMSO in concentration of 2 g/ml, 20 and 40 μ l were loaded on sterile discs under aseptic conditions and carefully placed onto pre-seeded medium inoculated with test pathogens as per McFarlands standard.

3. Results and discussion

3.1. XRD Analysis

The CuO/ZnO heterostructure's XRD pattern was analyzed using the X-ray diffraction (XRD) technique, as depicted in Fig. 1a. Distinct peaks were observed at specific 2θ angles: 31.7°, 34°, 36°, 47.5°, and 56.6°. These peaks were assigned to various crystal planes namely, (100), (002), (101), (102), and (110) of hexagonal ZnO, as identified by the JCPDS No. 36–1415 database. Additionally, other peaks were noted at 2θ angles of 16°, 38.7°, 49.8°, and 53°, corresponding respectively to crystal planes (100), (111), (202), and (020) of CuO. These findings align with the JCPDS number 45–0937 for CuO (tenorite), indicating the presence of a mixture of hexagonal and tetragonal CuO/ZnO with oxygen vacancies. Altogether, the XRD pattern furnishes valuable insights into the crystal

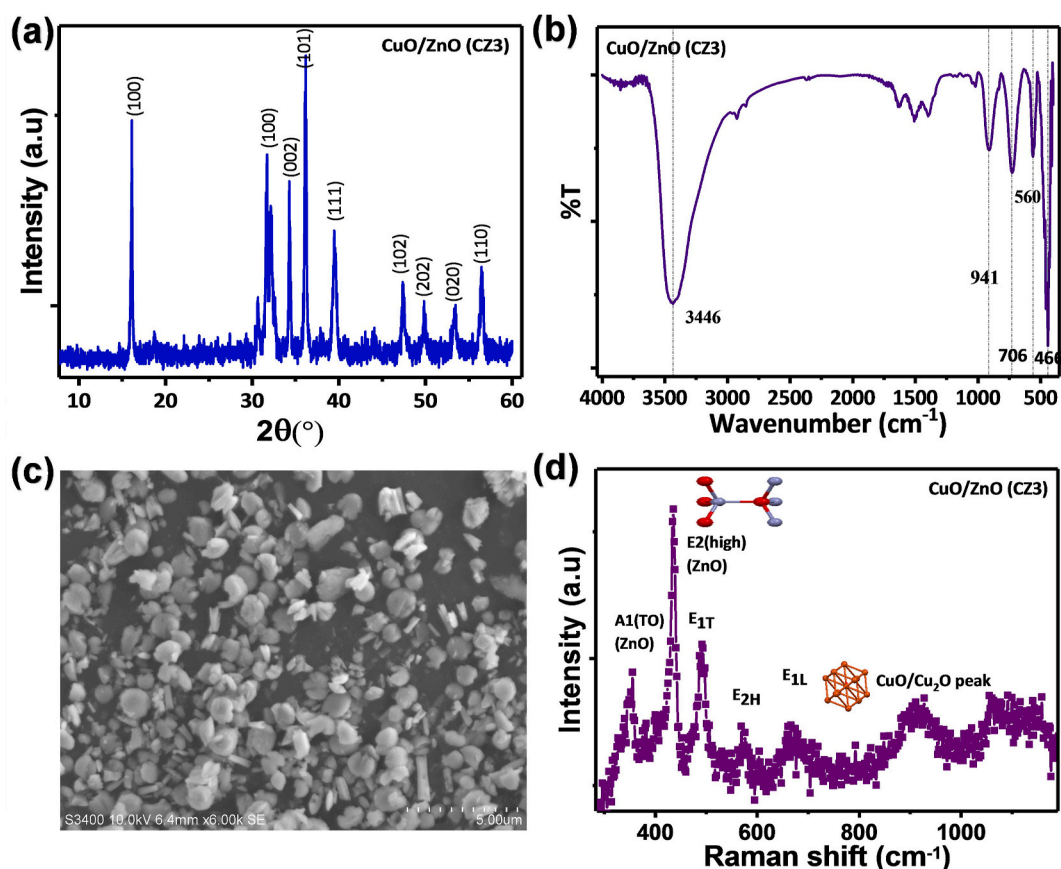


Fig. 1. a) XRD analysis of CuO/ZnO heterostructure, b) FTIR analysis of CuO/ZnO heterostructure, c) SEM image of CuO/ZnO heterostructure, d) Raman analysis of CuO/ZnO heterostructure.

structure and composition of the CuO/ZnO heterostructure.

3.2. FTIR spectral analysis

As shown in Fig. 1b, CuO/ZnO with abundant OH (3446 cm^{-1}) groups on its surface due to moisture and surface oxidation [47] can enhance its photocatalytic activity, as they can provide active sites for the adsorption of reactant molecules and facilitate the separation of photogenerated electron-hole pairs. The FTIR spectra of CuO/ZnO composite also showed a peak at 511 cm^{-1} , which is attributed to the stretching vibration of the Zn–O bond. This confirms the formation of ZnO in the composite material. The peak at 609 cm^{-1} corresponds to the stretching vibration of Cu–O bond, indicating the presence of CuO in the composite material. The presence of both CuO and ZnO in the composite creates a p-n junction, which enhances the separation of photogenerated electron-hole pairs and thus increases the photocatalytic efficiency of the composite.

3.3. SEM micrographs

The SEM analysis of CuO/ZnO composite provided valuable insights into the morphology and structure of the synthesized material. The image in Fig. 1c shows that the composite has a heterostructure with various morphologies. The bundle of wurtzite structure with hexagonal nanorods represents the CuO/ZnO heterostructure. The hexagonal nanorods of ZnO provide a large surface area for catalytic activity, while the CuO/ZnO heterojunction helps to enhance the photocatalytic performance. Overall, the SEM analysis confirms the successful synthesis of CuO/ZnO composite with desirable morphology and heterostructure.

EDX analysis is a powerful technique that determines the elemental composition of a sample by analyzing X-rays generated when the sample is bombarded with an electron beam. In this study, we have analyzed the SEM-EDX spectral elemental composition of the CuO/ZnO composite. Fig. 4d show the EDX spectrum that confirms the presence of three elements, namely Zn, O, and Cu, indicating the composite is composed of these elements. The presence of these elements confirms the high purity of the synthesized composite.

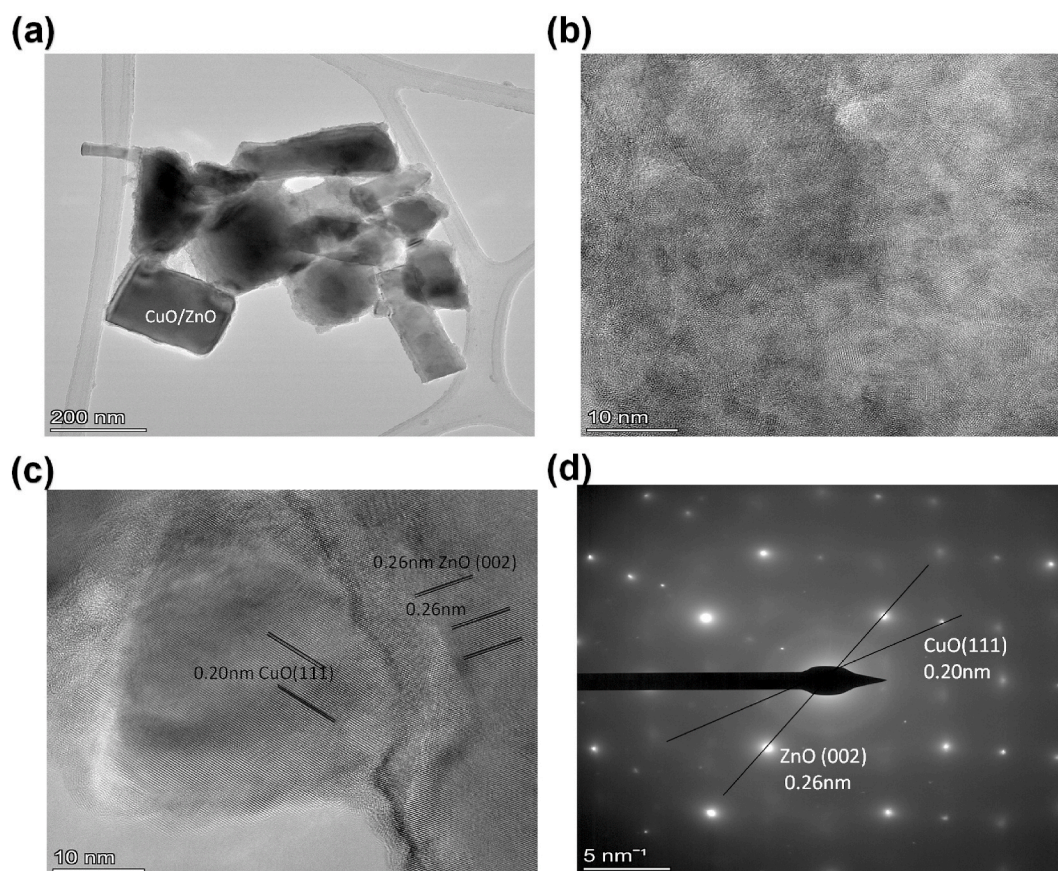


Fig. 2. CuO/ZnO characterization by TEM: a) TEM image of CuO/ZnO, b) and c) HRTEM images of CuO/ZnO, d) FFT analysis of HRTEM image of CuO/ZnO.

3.4. Raman spectroscopy

The Raman spectroscopy technique is a non-destructive method that is commonly used to analyze the properties of materials, including their chemical composition and structural information. As shown, in Fig. 1d, the synthesized CuO/ZnO composite was analyzed by Raman spectrometer, which revealed some important findings. The presence of the ZnO A_1 peak at 350/cm indicates the existence of the ZnO component in the composite. The positive shift observed in the 490/cm peak, which corresponds to the phonon of CuO/ZnO, compared to the ZnO peak at 424/cm, suggests a coupling between the CuO and ZnO components in the composite. This coupling is further confirmed by the presence of the E_2 peak at 490/cm, which is attributed to the CuO/ZnO composite. The peaks observed at 434/cm and 572/cm are related to the E_{1H} of CuO/ZnO and the E_{1L} of ZnO, respectively. These peaks provide information about the crystal symmetry and the vibration modes of the materials in the composite. The Raman spectra analysis of the CuO/ZnO composite provides important information about the properties of the material, which is useful for understanding the structure and composition of the composite.

3.5. TEM analysis

The TEM image of CuO/ZnO composite shown at Fig. 2, As shown in Fig. 2a, CuO (0.20 nm, (111)) nanoparticles are dispersed within a ZnO (0.26 nm, (002)) matrix. The dark spots in the image correspond to the CuO nanoparticles, while the brighter regions correspond to the ZnO matrix. The HRTEM image shows a high-resolution view of the interface between a CuO nanoparticle and the ZnO matrix in Fig. 2b. The lattice fringes visible in the image correspond to the crystal planes of CuO/ZnO. The image suggests that the CuO nanoparticles are well dispersed within the ZnO matrix and that there is a good interface between the two materials. Fig. 2c The HRTEM image at a different magnification provides additional details of the CuO/ZnO interface. The crystal planes of both CuO and ZnO are clearly visible, and the interface between the two materials appears to be well-defined. Fig. 2d display's The Fast Fourier Transform (FFT), it is a mathematical tool that can be used to analyze the periodicity of the crystal lattice in the HRTEM image. The FFT pattern provides information about the crystal structure and orientation of the materials in the image. In this case, the FFT pattern shows that both CuO and ZnO have a hexagonal crystal structure, consistent with their known crystal structures. The FFT pattern also indicates that the CuO nanoparticles are well dispersed within the ZnO matrix, and that there is a good interface between the two

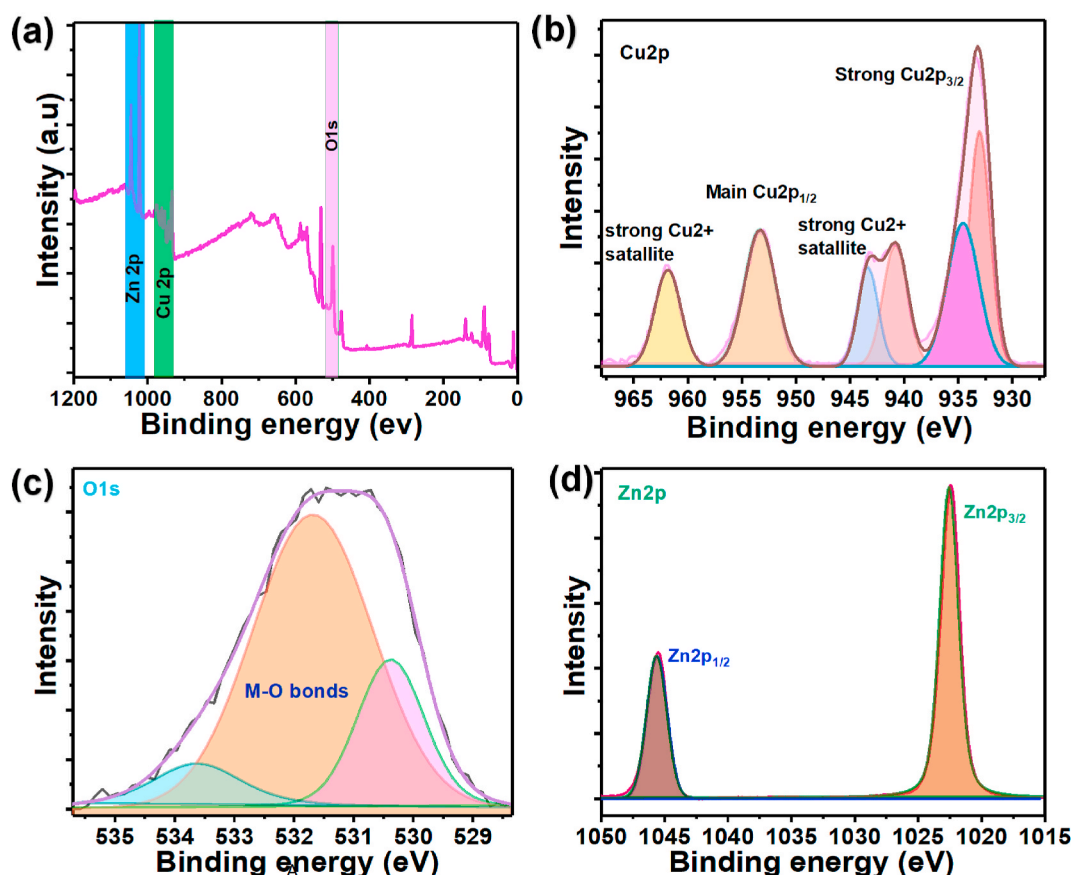


Fig. 3. CuO/ZnO (CZ3) XPS spectra a) Survey spectra of CuO/ZnO, b) Cu 2p spectra, c) O 1s spectra, d) Zn 2p spectra.

materials. The utilization of Energy Dispersive X-ray (TEM-EDX) analysis, a robust method, enables the identification of an object's elemental constitution through examination of X-rays generated upon electron beam bombardment. In this investigation, The EDX spectrum in Fig. 4c validates the existence of Zn, O, and Cu elements composition.

3.6. XPS analysis

The X-ray photoelectron spectroscopy (XPS) technique was employed in the present research work to investigate the binding energy of the CuO/ZnO composite and to determine the oxidation state of metals and oxygen vacancies in the photocatalyst. The XPS survey spectra of the CuO/ZnO composite were shown in Fig. 3a, which demonstrated the presence of copper and zinc, oxygen in the composite. In Fig. 3b, the high-resolution Cu 2p spectrum of the composite was shown, which allowed for the calculation of the oxidation state of copper. The satellite feature at 943 eV corresponded to Cu(II), while a weak satellite feature at 945 eV was related to Cu(I) oxide. Fig. 3c displayed the O1s peak, which provided information on the oxygen vacancies present in the photocatalyst. Finally, Fig. 3d showed the Zn 2p peak, which was split into two spin-orbit-related peaks, Zn 2p_{1/2} and Zn 2p_{3/2}. Overall, the XPS analysis provided valuable insights into the chemical composition and oxidation states of the CuO/ZnO composite, which can be useful for understanding its photocatalytic properties.

3.7. Optical bandgap study

The synthesized CuO/ZnO nanocomposite was analyzed by UV Visible spectrophotometer by light absorption and the Tauc plot were plotted using equation $(\alpha h\nu)^n = K(h\nu - E_g)$. Where α is the coefficient absorption, $h\nu$ is incident energy of the photon, K is constant, E_g is the energy gap, and n is the nature of transition, $n = 2$ for direct allowed transition, $n = 1/2$ for indirect allowed transition. Fig. 4a, shows the absorption spectra and tauc plots for CuO/ZnO photocatalyst in the wavelength region 200–800 nm at room temperature. The light absorption was measured at wavelength 554 nm which shows 2.4 eV bandgap. The above observations indicates that the CuO/ZnO have direct bandgap semiconductor properties in which Cu shows two electron transitions. In direct bandgap semiconductors the potential energy will change during transition from valence band to conduction band. In direct bandgap semiconductors, the radioactive recombination efficiency is high and degradation performance is high. The increase in the light absorbance capacity with direct band gap, increases the concentration of OH radicals and O²⁻ radicals for the high efficiency of composite.

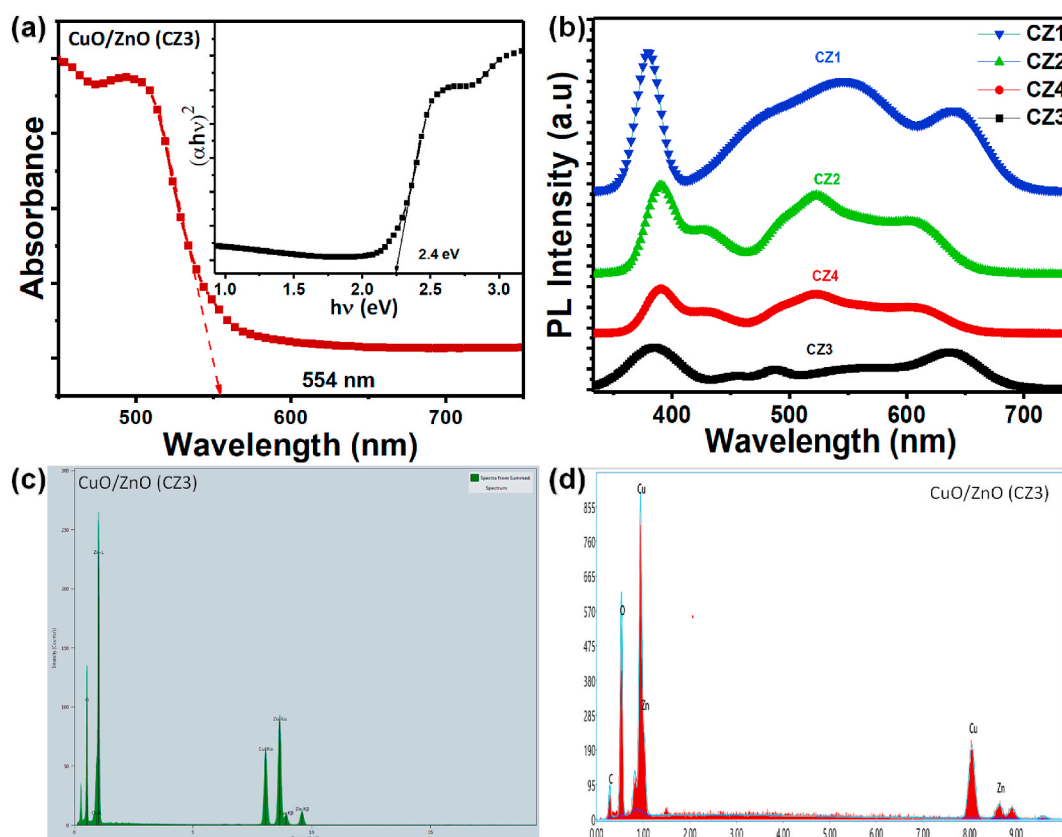


Fig. 4. a) UV-Vis spectral analysis along with Tauc plot, b) PL spectra, c) TEM-EDX spectra, d) SEM-EDX spectra.

3.8. Photoluminescence spectra

The photoluminescence spectra analysis of the CuO/ZnO heterostructure provides additional insight into the properties of the composite material. Fig. 4b shows the level of recombination and electron transition of the heterostructure, with the maximum PL intensity corresponding to maximum recombination of hole electron pairs. The PL spectra of CZ1, CZ2, CZ3, and CZ4 are shown in Fig. 4b, and the order of intensity is CZ2 > CZ1 > CZ4 > CZ3. This suggests that the CZ2 composite has the highest level of recombination and electron transition, while CZ3 has the lowest, which involves charge separation. It's interesting to note that the CZ3 composite has a direct bandgap of 2.4 eV. The decrease in recombination rate is observed with an increase in distance from the interface between the conduction band (CB) and holes of the CuO/ZnO composite semiconductor. These findings provide crucial insights into the behavior of electrons and holes within the CuO/ZnO heterostructure, which can aid in designing materials with desired optical properties.

3.9. Photocatalytic dye degradation

The methyl orange (MO) dyes have been found to be toxic and non-biodegradable at low concentrations, making it important to investigate their degradation using advanced techniques [48]. In this experiment, the degradation of MO was studied using CuO/ZnO photocatalyst under visible light. Fig. 5a shows the adsorption curve of MO with different weight percentages of CuO/ZnO photocatalyst as a function of time. The results indicate that the CZ3 photocatalyst exhibits the highest efficiency for MO degradation. A photocatalyst with 3 % Cu content was found to be optimal for dye degradation, as beyond this limit, the efficiency decreases. The initial MO concentration was 10 ppm, and the catalyst dosage was 0.5 g/L, with the pH set to 7. The study also analyzed the time-dependent adsorption (in the dark) and degradation of MO by different samples under visible light illumination as shown in Fig. 5b. The use of CuO/ZnO photocatalyst under visible light appears to be a promising approach for the degradation of MO, with the CZ3 photocatalyst showing the highest efficiency. This information could be useful for developing effective strategies for the removal of toxic dyes from the environment.

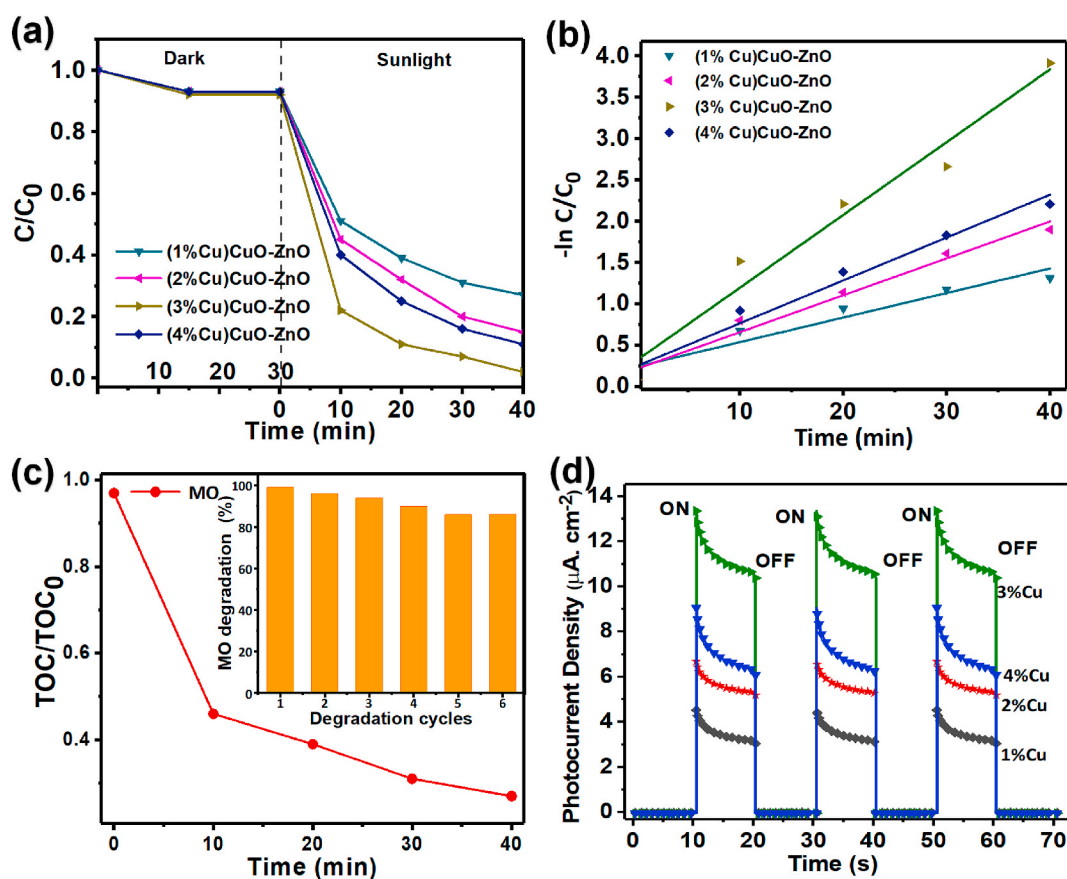


Fig. 5. CuO/ZnO photocatalytic performance: a) and b) Photocatalytic dye degradation of MO, c) CuO/ZnO Catalyst reusability and Mineralization degree d) Photocurrent measurements.

3.10. Reusability

To determine the reusability and stability of the CuO/ZnO photocatalyst, six consecutive experiments were conducted, and the efficiency of the CuO/ZnO was investigated. Fig. 5c shows the reusability of the photocatalyst CZ3 for six cycles, with the efficiency of the first cycle being approximately 99 %, while that of the sixth cycle was about 89 %. The results indicate that the photocatalyst CZ3 is highly efficient and stable, after six cycles of use, there was only a slight reduction in efficiency. This suggests that the CuO/ZnO photocatalyst could be a promising candidate for practical applications that require repeated use, such as wastewater treatment. However, it is important to note that the efficiency of the photocatalyst may depend on various factors such as the concentration of the dye, the catalyst dosage, and the pH of the solution. Further studies are needed to optimize the performance of the photocatalyst under different conditions.

3.11. Mineralization degree and effect of catalyst dosage

The mineralization of degraded dyes was analyzed using TOC technique, and the results are presented in Fig. 5c. The mineralization process was calculated for the photocatalyst CZ3 during the degradation of MO dye. The degradation process was conducted using different weight percentages of photocatalyst ranging from 0.2 g/L to 0.6 g/L. However, the optimal concentration for dye degradation was found to be 0.5 g/L of photocatalyst. Any concentration beyond this led to a decrease in efficiency.

3.12. Photocurrent

Fig. 5d shows the photocurrent spectra analysis of the as-synthesized CuO/ZnO heterostructure. The intensity of the photocurrent was observed to remain constant under light, indicating the continuous generation of electron-hole pairs. However, when the light was turned off, the intensity of the photocurrent decreased to zero. The photocurrent generated by the individual CuO and ZnO components was found to be larger compared to that of the CuO/ZnO hybrid composite. This is attributed to the presence of an internal electric field in the CuO/ZnO heterostructure, which enhances the separation of photogenerated electron-hole pairs, resulting in a higher efficiency of photocatalysis. These findings suggest that the CuO/ZnO hybrid composite is a promising material for photocatalytic applications due to its high efficiency in generating photocurrents. The internal electric field present in the heterostructure facilitates the efficient separation of the photogenerated electron-hole pairs, resulting in enhanced photocatalytic performance.

3.13. Docking studies

As shown in Fig. 6a, b and 6c, the docking data are listed in Table 1; they showed that all target receptors had good binding energies. The maximum binding energy value for 1MOQ was discovered, showing a strong connection between the ligand and the docked proteins Fig. 6b. It was discovered that the ligand interacted with it in a Pi-Sigma, Pi-Donor, Charge-Charge, and Metal Acceptor manner. The molecular docking demonstrated that the nanocomposite was a strong candidate with biological potential due to its higher binding energy.

3.14. Anticancer activity (MTT assay)

As shown in Fig. 7. The anti-cancer activity of CuO/ZnO heterostructures were tested on prostate cancer cells (PC-3), and an IC₅₀ value were obtained by plotting non-linear regression curve fit. In this investigation, the CuO/ZnO heterostructures, when subjected to varying degrees of exposure Fig. 7a, b, 7c, 7d, and 7e, were found to inhibit the growth of PC3 cancer cells. Prescribed dosages used in the study revealed structural and cellular abnormalities, and induced apoptosis in the PC3 cells, whereas the control or untreated groups revealed no such changes Fig. 7a. These findings indicated, without a shadow of a doubt, that the pre-phase apoptotic cell counts rose as the concentrations of CuO and ZnO heterostructures increased. Because of the damage done to the cells, their structure

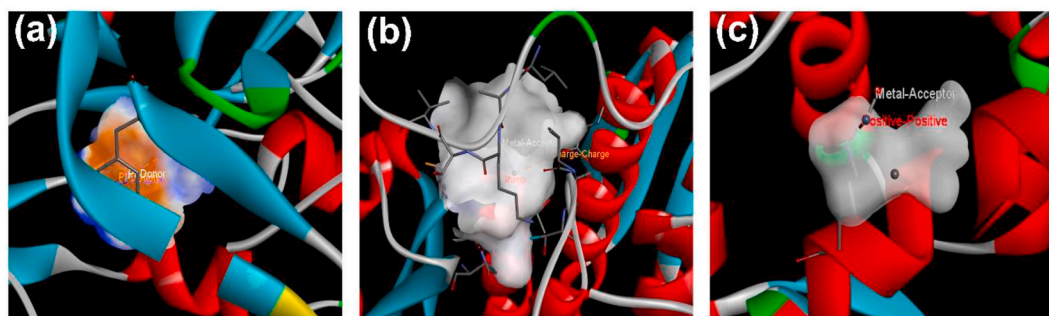


Fig. 6. a) Molecular Interaction of CuO/ZnO with 1IYL, b) Molecular Interaction of CuO/ZnO with 1MOQ, c) Molecular Interaction of CuO/ZnO with 3KP4.

Table 1
Molecular docking and its interactions.

PDB ID	Binding energy
1IYL	−5.01
1MOQ	−5.45
3KP4	−5.37

Table 2
Antibacterial zone of inhibition tested against various bacterial strains.

Organism	Positive control	Sample 20 μ l	Sample 40 μ l	Negative control
<i>Escherichia coli</i>	23.66 $\pm$ 0.33	35.66 $\pm$ 0.33	42.33 $\pm$ 0.33	Nil
<i>Staphylococcus aureus</i>	23.33 $\pm$ 0.33	32.33 $\pm$ 0.33	36.00 $\pm$ 0.33	Nil
<i>Bacillus cereus</i>	25.33 $\pm$ 0.33	32.66 $\pm$ 0.33	37.33 $\pm$ 0.33	Nil
<i>Pseudomonas aeruginosa</i>	22.66 $\pm$ 0.33	29.33 $\pm$ 0.33	34.66 $\pm$ 0.33	Nil

needed to be substantially altered. In general, metal oxide nanocomposites and heterostructures caused only a negligible amount of harm; nevertheless, these materials, after undergoing both modest and substantial modifications, have the potential to one day find use in the commercial sector. We study the impact of the bio-toxicity of the CuO/ZnO heterostructures on the cellular adherence of prostate cancer cells (PC-3 cells) in order to address the evidentiary gap that currently exists in the research on biocompatibility. The cells that had been treated with CuO/ZnO displayed characteristics of apoptosis, such as cell contraction, severely shortened and detached nuclei, the formation of a large number of apoptotic vesicles, in addition to the breakdown and degeneration of cells [49]. The lethal impacts of a large CuO/ZnO nanocomposite concentration led researchers to hypothesize that apoptosis was the principal channel via

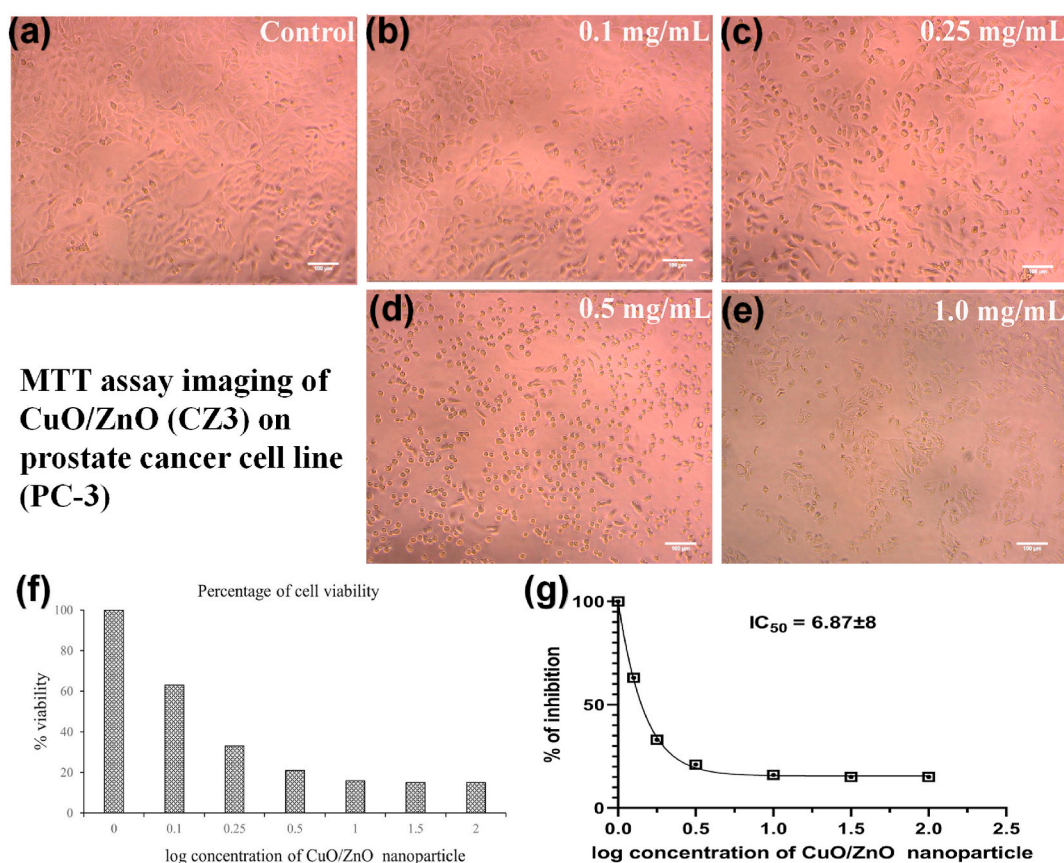


Fig. 7. Microscopic pictures of cells subjected to various concentrations of CuO/ZnO composite [a] control, b) 0.1, c) 0.25, d) 25, e) 0.50, and f) 1.0 Mg/ml, B) graph showing percentage viability versus concentration of CuO/ZnO composite using MTT assay, and C) graph for determining IC₅₀ value for CuO/ZnO composite.

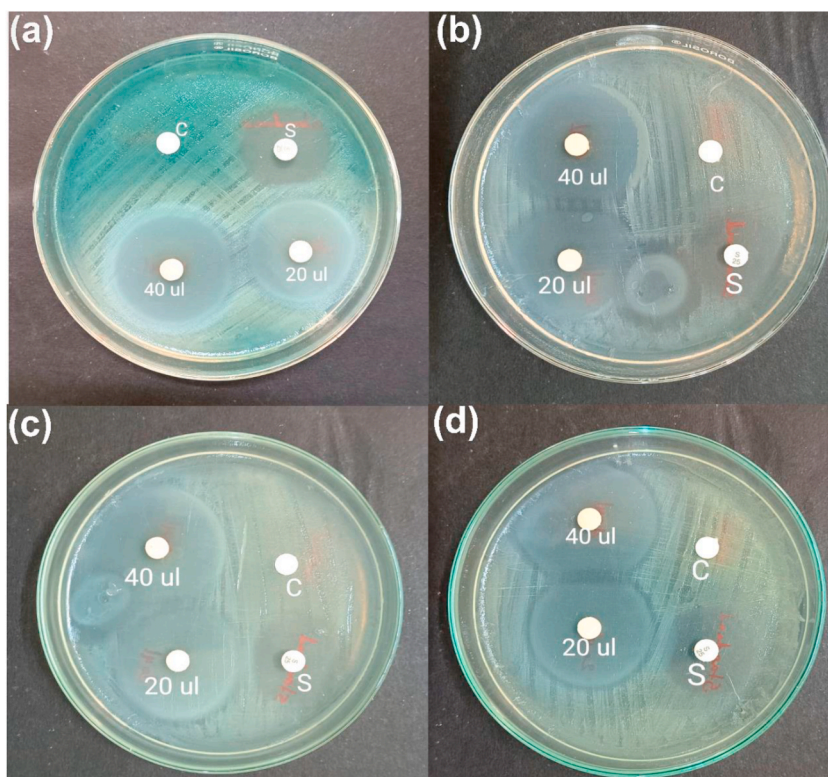


Fig. 8. The observations of a dose-dependent decrease in bacterial growth (mm) were made using varying concentrations of CuO/ZnO composite. The zone of inhibition (ZOI) that was observed around the well loaded with CuO/ZnO composite in plates that were seeded with a bacterial suspension of *Escherichia coli*, *Staphylococcus aureus*, *Bacillus cereus*, and *Pseudomonas aeruginosa* is shown in the inset. The data represent mean $\pm$ SD ($n = 3$).

which cells were directed towards apoptosis. The control dose was the globally acknowledged 50 % inhibitory dose determined using IC_{50} values. As indicated in Fig. 6f & g, the IC_{50} value was 6.87 ± 8 .

3.15. Antibacterial activity

To determine the antibacterial efficacy of nanocomposites, in the current study, As shown in Fig. 8. *Escherichia coli*, *Staphylococcus aureus*, *Bacillus cereus*, and *Pseudomonas aeruginosa* were chosen because of their ability to cause disease in human populations. Because of the broad spectrum of metabolism and action shared by both drugs, gentamicin and ciprofloxacin were traditionally used as the gold standards for antibiotic treatment 10 μ g was taken as standard [50]. The sample was subsequently tested using the disc diffusion method to determine their ability to inhibit test pathogenic bacteria. The results of the antibacterial activity against the test pathogens indicated considerable inhibition [51]. CuO/ZnO nanocomposites inhibited *Escherichia coli* the most (42.33 0.33 mm) Fig. 8a, *Staphylococcus aureus* the next (37.00 0.33 mm) Fig. 8b, *Bacillus cereus* the next (30 mm) Fig. 8c, *Staphylococcus aureus* the next (36.33 0.33 mm) Fig. 8b, and *Pseudomonas aeruginosa* the next (34.66 0.33 mm) Fig. 8d. As seen in Fig. 8., and Table 2, CuO/ZnO nanocomposites exhibited antibacterial action. When compared to gram-positive bacteria, the results showed that gram-negative pathogenic bacteria were highly susceptible to CuO/ZnO composite.

4. Conclusion

This research work focused on the synthesis of CuO/ZnO composite using a green technology-oriented microwave hydrothermal method. Various advanced analytical techniques were employed to analyze the synthesized composite, and its photocatalytic activity was investigated for the degradation of methyl orange (MO) dye. The results showed that the CuO/ZnO p-n junction photocatalyst had a high efficiency for charge transport and suppressed charge carrier recombination, resulting in improved photocatalytic activity. Molecular docking studies also demonstrated the potential of CuO/ZnO nanocomposites for different proteins, with promising anti-cancer and antibacterial properties. The overall findings suggest that the synthesized nanoparticles possess both photocatalytic and biocatalytic properties, with potential as a drug material. Further studies are necessary to fully understand and explore the underlying mechanisms of these phenomena and to optimize the synthesis of CuO/ZnO nanocomposites for various practical applications. Nonetheless, this research work provides a valuable contribution to the field of nanotechnology and photocatalysis.

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Data availability

The datasets used and/or analyzed during the current study available from the corresponding author on reasonable request.

CRediT authorship contribution statement

Deepu Habbanakuppe Ramakrishnegowda: Writing - review & editing, Writing - original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kampalapura S. Chandrakantha:** Writing - review & editing, Writing - original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Deepadarshan Urs:** Writing - review & editing, Visualization, Methodology, Formal analysis. **Mohamed Elfeky:** Writing - review & editing, Conceptualization. **Jagadish Krishnegowda:** Writing - review & editing, Formal analysis. **Shobith Rangappa:** Data curation, Conceptualization. **Kanchugarakoppal S. Rangappa:** Supervision, Project administration, Investigation, Conceptualization. **Srikantaswamy Shivanna:** Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

We would like to submit an original research article to the Heliyon from Elsevier with the title "Synthesis of p-CuO/n-ZnO Heterostructure by Microwave Hydrothermal Method and Evaluation of Its Photo and Bio-Catalytic Performance." This work is authentic, unique, and not currently being considered for publication anywhere else, as we have confirmed. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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